Nirbhay Katiyar

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Software Developer focused on building secure, scalable applications. Experienced in Building automated pipelines, load-heavy systems, cloud-native deployments.

Skills

Languages: Python, Go, C++
Backend & APIs:FastAPI, REST APIs, gRPC, PostgreSQL, DynamoDB, REGEX
AI/ML:LLM Integrations, AI Agents, Agentic Workflows, RAG, LLM Orchestration, MCP, Vector DBs, SLM
Cloud & DevOps: AWS (Lambda, EC2, CloudWatch, Bedrock, EventBridge, S3, DynamoDB), Docker, Serverless, Microservices, CI/CD
Security: OWASP Top 10, Penetration Testing, JWT Hardening, SQLi, XSS, CSRF, SSRF, CORS, Burp Suite, NMAP, PCAP, Wireshark
CS Fundamentals: Operating Systems, Computer Networking, DBMS, TCP-IP, System Design, OOP, LLD, DSA

Experience

Backend & Operations Engineer Aug 2026 – Present
Plany *Remote*

- Migrated the platform from InsForge to AWS with zero downtime, designing the cloud layout across services, data, networking, and deployment for clearer ownership and room to scale.
- Integrated AI capabilities into the product's backend services and built the authentication layer for the platform.
- Optimized infrastructure usage and resource allocation across services, cutting cloud costs by 37% while maintaining reliability.

Independent Security Researcher (Bug Bounty) Feb 2024 – Jan 2025
HackerOne *Remote*

- Responsibly disclosed critical vulnerabilities in platforms operated by Spotify, LightSpark, and IBM, earning valid reports.
- Researched attack chains involving IDOR, auth bypass, SSRF, and information disclosure with clear business impact.

Projects

ZeroDaily | *AI Tech News App* | *Python, FastAPI, AWS Lambda, DynamoDB, Bedrock, React Native* Live Website

- Built a serverless news platform that ingests ∼2,500 articles/day from 24 RSS feeds every 8 minutes and publishes ∼400 LLM-summarized stories/day to a React Native mobile app.
- Cut scraping and inference cost by ∼75% via LLM semantic deduplication of headlines before any fetch; Bedrock with API fallback, circuit breakers, and backoff.
- Hardened the pipeline with SSRF-guarded fetching, prompt-injection isolation of article text, and schema-bounded outputs.
- Delivered FCM topic-based push with DynamoDB conditional-write cooldowns and Cloudflare cache warming; API responds in 100–400ms under k6 load, backed by 172 tests and gated deploys.

FlexURL | *Production URL Management Platform* | *Python, FastAPI, PostgreSQL, AWS* Live Website

- Architected a URL platform with Snowflake IDs (Base62), serving 410 redirects/sec at 100% success on a 2vCPU server.
- Built async analytics with ARQ, keeping redirect p95 latency below 210ms (avg 42ms).
- Engineered custom domains, OS-aware redirection, password protection, expirations, QR codes, REST APIs, and RBAC.
- Built a real-time analytics pipeline with webhooks, tracking geo-location and device metrics, using Redis to minimize DB load.
- Hardened with SSRF/CSRF mitigation, Google Safe Browsing, token-bucket rate limiting, and Cloudflare edge protection.

Conduit | *Local-First Browser Control Bridge for AI Agents* | *TypeScript, Node.js, MCP* Live Website

- Built an open-source bridge (daemon, CLI, MCP server, and Manifest V3 extension) that lets AI agents operate a user's real signed-in browser, live on the Chrome Web Store and Firefox Add-ons.
- Secured remote access with P-256 proof-of-possession, per-request signed nonces, revocable device keys, loopback-only binding, and rate-limited authentication.
- Enforced least privilege via dual authority gates (per-site host permissions and daemon capability/domain policy) and one-time, 60-second confirmations for high-risk actions, backed by a redacting audit log.

Open Source Contributions

LibrePhotos (Python, Django, React): Added a "Without Location" view for photos missing GPS data, with tests. Merged.
LibrePhotos (Python, Django): Server-side "select all" so bulk actions cover every matching photo. Merged.
Glaze (C++): Fixed `operator[]` on generic JSON to accept integer variables, not just literals, with lvalue/rvalue tests. Merged.

Education

Galgotias College of Engineering & Technology Greater Noida, UP
B.Tech in Electronics & Communication Engineering (Expected Apr 2029) *CGPA: 7.59*